Inverse Transition Learning from Expert Demonstrations:
Reproduction and a Joint Reward–Dynamics Extension

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1 Motivation and question

Standard reinforcement learning knows the dynamics T of a Markov decision process and wants the optimal policy π^*. Many real settings invert that assumption: [1] uses an intensive care unit as its running example, where outcomes are observable but how a patient responds to a particular dose is not, and what we see is a batch of clinician decisions that are close to optimal but not exactly optimal.

My broader interest sits downstream of this in personalized wellness technology: the wearable and continuous-monitoring stack (Apple Watch, Oura, WHOOP, continuous glucose monitors). The dominant industry pattern is to fit a supervised model from user action to outcome, treating every body as a shared-dynamics black box and prescribing one-size-fits-all behavioral interventions. Inverse Transition Learning (ITL) takes a different posture: the user is a near-optimal decision-maker whose actions encode information about their own physiology, and that signal is exactly what maximum-likelihood-over-counts throws away. Recovering individualized dynamics from those actions is the methodological gap I want to work in. The semester's technical question: given the reward R and a batch of ε-optimal expert transitions $D = \{(s, a, s')\}$, can we recover T^* in a way that respects the optimality information in the actions?

2 Method and tools

An action a is ε-optimal at state s when $\max_{a'} Q^*(s, a'; T) - Q^*(s, a; T) \leq \varepsilon$. Observing the expert choose a at s yields an inequality on Q^*. Writing the policy-evaluation inverse $v^\pi = (I - \gamma P^\pi)^{-1} r^\pi$ and substituting the Laplace-smoothed MLE T^{MLE} inside it linearizes the constraint in T, giving

$$\min_T \ \frac{1}{2} \sum_{s,a,s'} N_{s,a,s'} \big(T_{s,a,s'} - T^{\text{MLE}}_{s,a,s'}\big)^2 \quad \text{s.t.} \quad \varepsilon\text{-ball on } (s,a) \in D, \text{ simplex on } T.$$

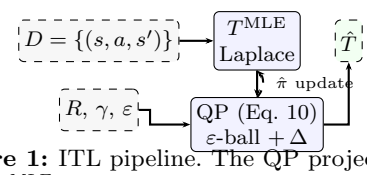

Figure 1: ITL pipeline. The QP projects T toward T^{MLE} subject to ε-optimality of the expert's observed actions; $\hat{\pi}$ and the QP alternate (Algorithm 1) until convergence.

The solver alternates between a policy estimate $\hat{\pi}$ (uniform over observed actions; uniform over all actions at unobserved states) and the QP, terminating when the ε-ball holds at every observed state (Fig. 1). The convex pieces map onto Krause and Hübotter [2, Ch. 10–11]: Bellman optimality, the linear policy-evaluation system, and the Dirichlet–Categorical conjugacy that makes the Laplace-smoothed MLE a posterior mean rather than a frequentist estimate. Implementation: CVXPY [5] with an OSQP–SCS–CLARABEL fallback chain.

3 Simulation: coverage sweep on Gridworld

I reproduced the paper's central empirical claim (their Figure 2) on a 5×5 Gridworld with soft walls (penalty -5), slip 0.2, $\gamma = 0.95$, $\varepsilon = 5.0$, and a 40%-stochastic expert. *Coverage* is the fraction of states appearing in D; for each observed (s, a) I sampled 10 transitions. At full coverage, ITL achieves Normalized Value 0.998 ± 0.001 versus MLE's 0.142 ± 0.002, with 0.00 versus 7.20 constraint violations and ε-matching 1.000 versus 0.712 (5 seeds; Fig. 2). MLE

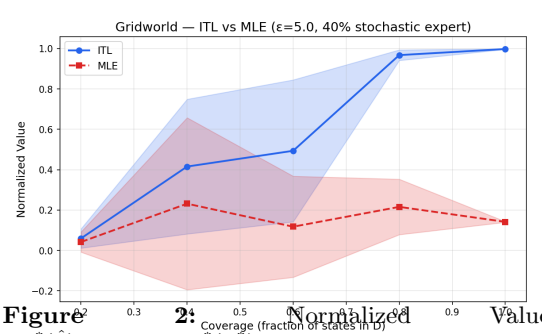

Figure 2: Normalized Value $V^{\pi^*(\hat{T})}(s_0; T^*)/V^{\pi^*(T^*)}(s_0; T^*)$ vs. coverage on Gridworld. ITL plateaus near 1.0 at full coverage and dominates MLE everywhere. Bands: ±1 std.

under-counts visited next-state cells when the expert is occasionally stochastic, so its greedy policy walks into soft walls; ITL avoids this by enforcing that the expert's observed choices remain ε-optimal under $\hat{T}$. The shape matches the paper's Figure 2; the absolute scale runs $\sim$ 1.5$\times$ larger than Table 4, almost certainly because the soft-wall layout is not detailed in the paper, so my environment is somewhat less aversive. The structural pattern (ITL $\geq$ MLE on every metric, ε-matching $\rightarrow$ 1.0 at full coverage, per Theorem 1) holds on both Gridworld and RandomWorld.

4 Open question: joint reward and dynamics recovery

The paper assumes R is known. In MIMIC-IV that is fragile: the survival proxy reward ($\pm$15 for discharge or mortality, after [4]) is hand-engineered. The extension I have been prototyping, named in the paper's future-work section, lifts R into a decision variable $R(s,a) = \Phi(s,a)^\top w$ and solves for (T, w) jointly (Fig. 3). The ε-ball constraint becomes

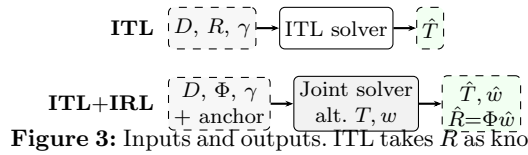

Figure 3: Inputs and outputs. ITL takes R as known input; the extension takes a feature map Φ plus a single anchor and recovers $(\hat{T}, \hat{w})$ jointly, returning a learned reward $\hat{R} = \Phi\hat{w}$.

$$\big(\Phi(s,a) - \Phi(s,a')\big)^\top w + \gamma\,(T_{s,a} - T_{s,a'})^\top v_{\text{lin}} \;\geq\; \varepsilon,$$

bilinear in (T, w) but a convex QP at each step of the alternation (hold w, solve for T; hold T, solve for w). An L1 penalty on w plus an anchor at one (s,a) pin the additive ambiguity. The prototype converges to a fixed point in two outer iterations on both the 3-state corridor and the 5$\times$5 Gridworld. On Gridworld with one-hot (s,a) features ($d = 100$), goal-anchored at $R(s^*, 0) = 10$, $\lambda_{L1} = 0.01$, deterministic expert: ITL+IRL recovers the optimal policy exactly (best-action 1.000 vs. ITL with known R 1.000 vs. MLE 0.600); T MSE 0.4873 vs. 0.4929 (ITL) vs. 0.5154 (MLE). Remaining before May 31: a head-to-head against MaxEnt-IRL [3] (now wired in via L-BFGS-B with a null-space-parameterized anchor) under stochastic experts and coarser feature maps where identifiability bites.

Implications for personalized wellness. The reproduction confirms that when behavior is near-optimal, individual-level dynamics are recoverable from sparse demonstrations in a way supervised models miss. The joint reward-and-dynamics extension matters more for wearables than for the ICU: sleep-quality or recovery scores are noisy user proxies, not ground truth, so inferring R jointly with T removes the most fragile assumption in any wearables pipeline. The continuous-state lift is the gating step before this transfers to raw sensor streams.

Code and reproducibility. Reproduction codebase (Python 3.10, CVXPY, MIT) at `https://github.com/REPLACE-ME/itl-reproduction`, with inline cross-references to paper equations and textbook chapters. A Colab that re-runs the corridor sanity check, the Gridworld coverage sweep at reduced seed count, and the ITL+IRL smoke test is at `https://colab.research.google.com/REPLACE-ME`.

References

References

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